

Employee Performance & Payroll Analysis

A complete data cleaning, analysis & dashboard project

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100 Employees | 5 Departments | 3 Job Levels



100

Total Employees



\$9.94M

Total Payroll



3.63 / 5

Avg Performance



\$89,330

Avg Salary

PROJECT OVERVIEW

What is this project all about?

This project focuses on analyzing a real-world style HR dataset covering 100 employees across 5 departments. The goal was to clean messy raw data and extract meaningful insights.

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1

Data Exploration

Studied raw data — checked for nulls, inconsistent text, wrong formats, and missing values

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Data Cleaning

Fixed all 6 issues: null values, text case, extra spaces, date formats, duplicates & data types

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Pivot Analysis

Built pivot tables to summarize salary, bonus, performance by department and job level

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Dashboard

Created a visual dashboard with KPI cards, charts, and tables for easy decision making

DATA CLEANING PROCESS

SIX Key Cleaning Tasks Applied on Raw Data

Task 1: Handled Null / Missing Values

Filled missing salary & bonus fields with department averages — no blank cells left in dataset

Task 2: Standardized Text Case

Converted all Department and Job Level columns to proper Title Case (e.g., 'Employess_Salary' → 'Employess_Bonas_Salary')

Task 3: Removed Extra Spaces

Trimmed leading/trailing spaces from Employee_ID, Department, and Job_Level columns

Task 4: Corrected Date Formats

Unified all date fields to Year format for consistent sorting and filtering

Task 5: Removed Duplicate Rows

Identified and removed duplicate Employee_ID entries — ensuring data integrity

Task 6: Fixed Data Types

Converted Salary to Number, Bonus_Percentage to Decimal, and Rating to Integer type

Result: Clean, structured, analysis-ready dataset

KEY METRICS & KPIs

100

Total Employees

Across 5 departments

\$9.94M

Total Payroll

Annual total cost

\$1.01M

Total Bonus Paid

10.2% of payroll

3.63★

Avg Performance

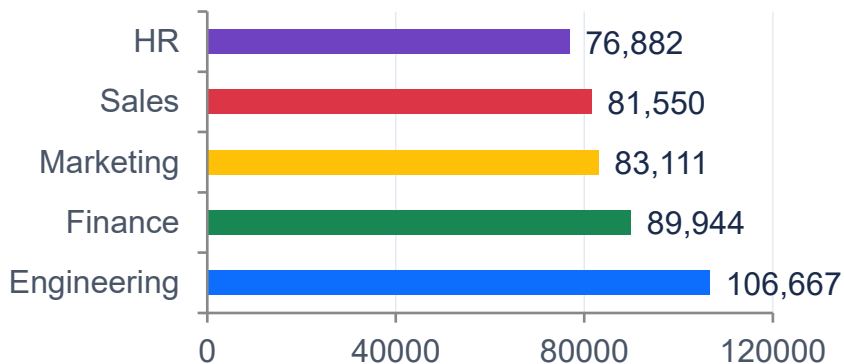
Out of 5.0

6.0 yrs

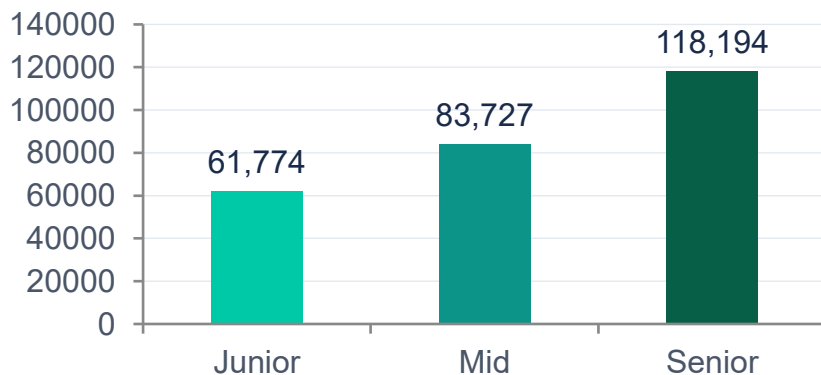
Avg Experience

Company average

Average Salary by Department

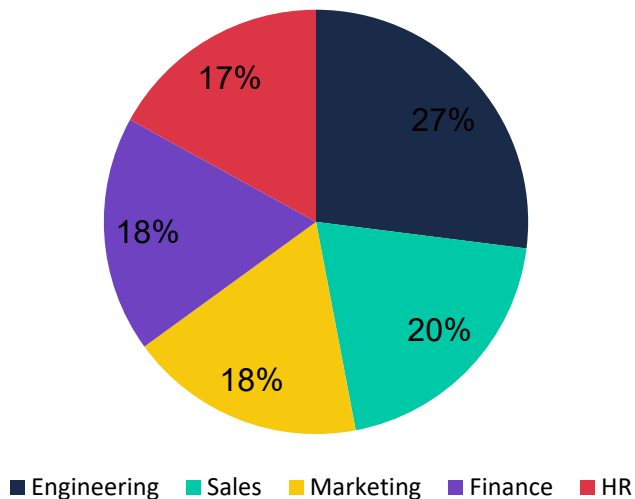


Avg Salary by Job Level

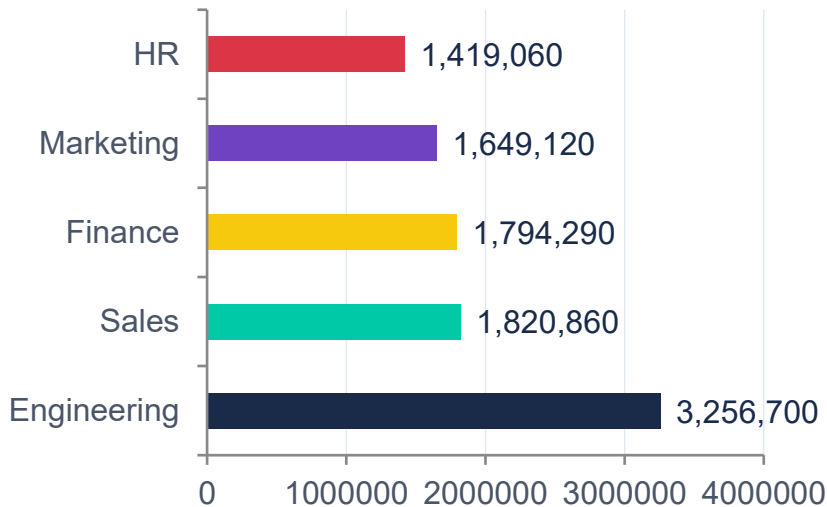


DEPARTMENT-WISE ANALYSIS

Employee Count by Department



Total Payroll by Department (\$)

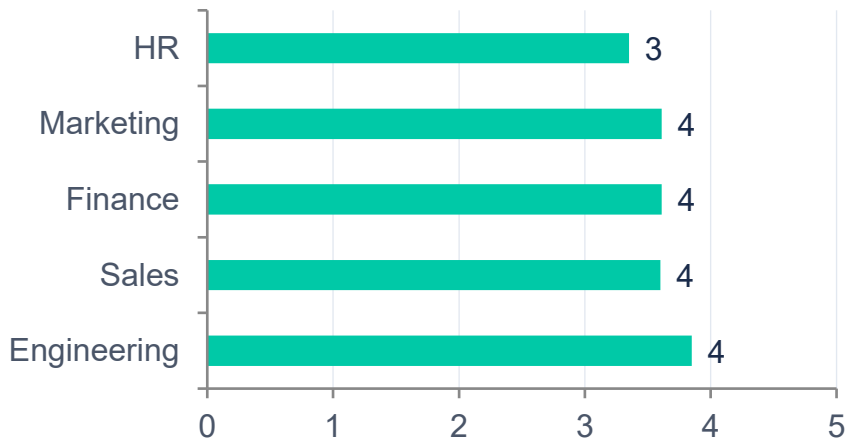


Key Insights:

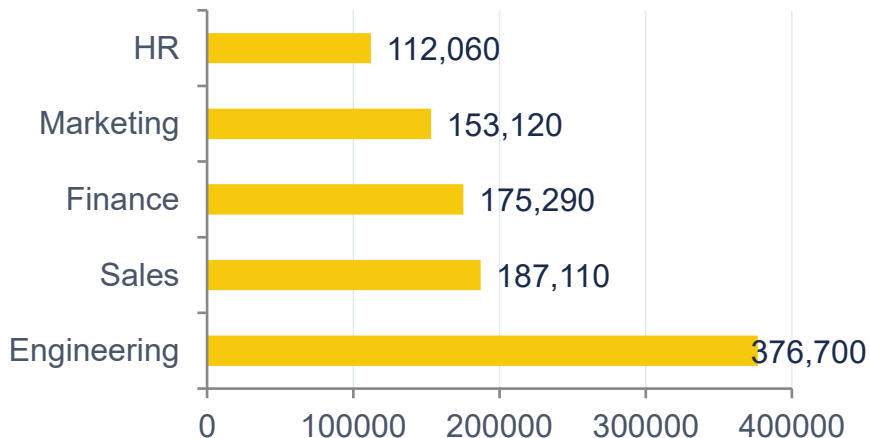
- Engineering is the largest dept (27%) & highest paid — avg \$106,667/year
- HR has the fewest employees (17) and the lowest average salary (\$76,882)
- Engineering consumes 33% of total payroll — \$3.25M out of \$9.94M

PERFORMANCE & BONUS INSIGHTS

Avg Performance Rating by Department



Total Bonus Paid by Department (\$)



What the numbers tell us:

- Engineering leads in both performance (3.85 avg rating) and total bonus (\$376K)
- Employees with Rating 5 earn on average \$136,917 in total — vs \$72,803 for Rating 2
- Senior employees receive up to 25% bonus — Juniors often get 0% for low performance ratings
- HR has the lowest avg performance rating (3.35) — potential area for improvement

Key Takeaways & Summary

- 1 Successfully cleaned 100 employee records — handled nulls, duplicates, case issues & formats
- 2 Engineering dept has the highest salary, headcount (27%), and performance scores
- 3 Clear link between performance rating and salary — higher rating = significantly higher pay
- 4 Bonus structure rewards Senior-level high performers the most (up to 25%)
- 5 Dashboard provides management with quick visual access to all HR KPIs at a glance

Thank You!

Project Stats

100

Employees Analyzed

5

Departments Covered

6

Cleaning Tasks Done

4

Charts in Dashboard

**\$9.9
4M**

Total Payroll